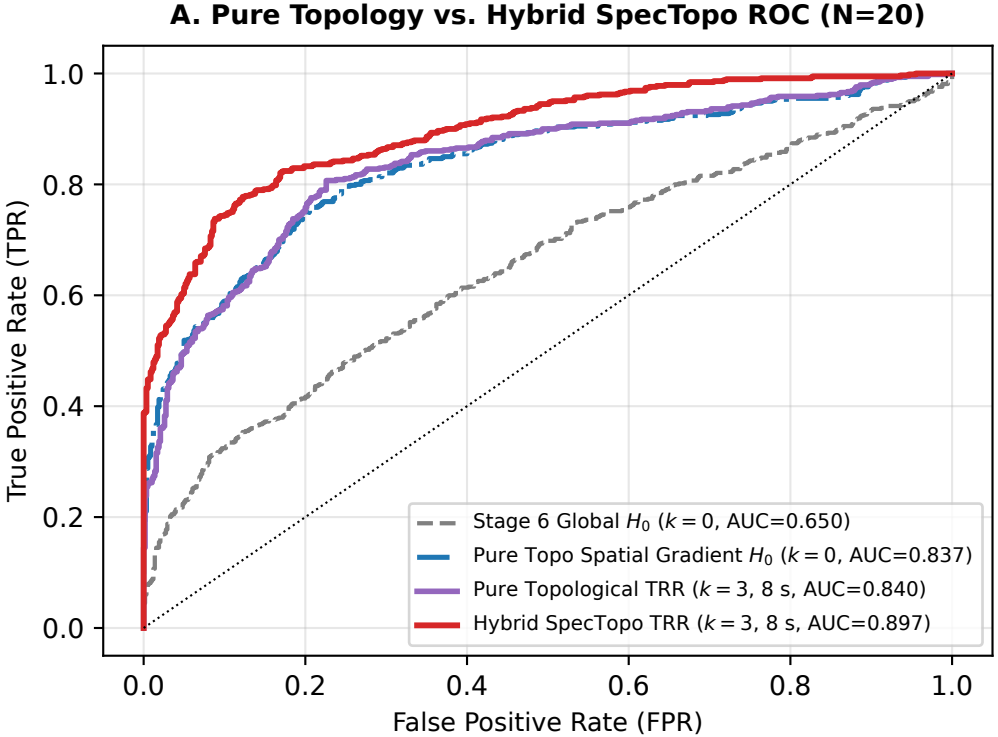
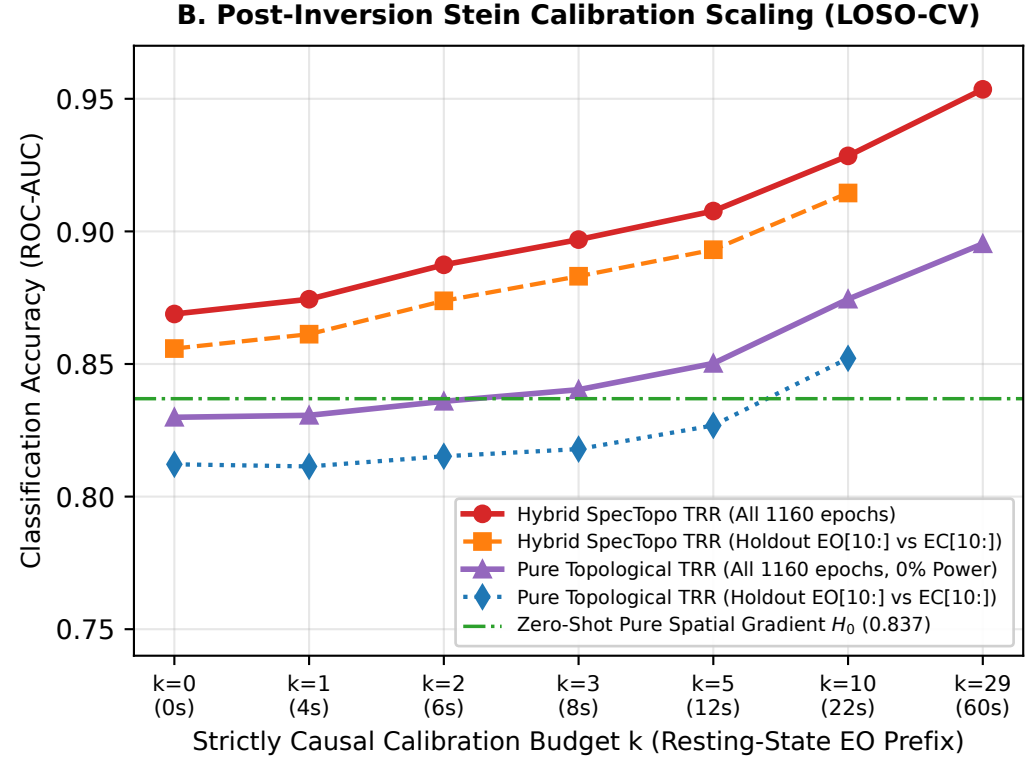

A. Pure Topology vs. Hybrid SpecTopo ROC (N=20)
True Positive Rate (TPR)
False Positive Rate (FPR)
0.0
0.2
0.4
0.6
0.8
1.0
Stage 6 Global H_0 ($k = 0$, AUC=0.650)
Pure Topo Spatial Gradient H_0 ($k = 0$, AUC=0.837)
Pure Topological TRR ($k = 3$, 8 s, AUC=0.840)
Hybrid SpecTopo TRR ($k = 3$, 8 s, AUC=0.897)


B. Post-Inversion Stein Calibration Scaling (LOSO-CV)
Classification Accuracy (ROC-AUC)
0.75
0.80
0.85
0.90
0.95
k=0 (0s)
k=1 (4s)
k=2 (6s)
k=3 (8s)
k=5 (12s)
k=10 (22s)
k=29 (60s)
Strictly Causal Calibration Budget k (Resting-State EO Prefix)
Hybrid SpecTopo TRR (All 1160 epochs)
Hybrid SpecTopo TRR (Holdout EO[10:] vs EC[10:])
Pure Topological TRR (All 1160 epochs, 0% Power)
Pure Topological TRR (Holdout EO[10:] vs EC[10:])
Zero-Shot Pure Spatial Gradient H_0 (0.837)